

Application Deployment

(Deploy the given React application to a production ready state)

Application:

Clone the below mentioned repository and deploy the application (Run application in port 3000).

Repo URL : <https://github.com/Vennilavan12/Brain-Tasks-App.git>

Docker:

- Dockerize the application by creating Dockerfile
- Build an application and check output using docker image.

ECR:

- Create an AWS ECR repository for store docker images.

Kubernetes:

- Setup Kubernetes in AWS EKS and Confirm EKS cluster is running.
- Write deployment and service YAML files.
- Deploy using kubectl via Codedeploy.

CodeBuild:

- Create a CodeBuild project:
- Source: Connect to your repository
- Environment: Use managed image (Amazon Linux, Ubuntu)
- Write and define commands in buildspec.yml.

CodeDeploy:

- Create codedeploy application.
- create appspec.yml file to deploy applications in EKS.

Version Control:

- Push the codebase to a Git provider (GitHub).
- Use CLI commands to push code.

CodePipeline:

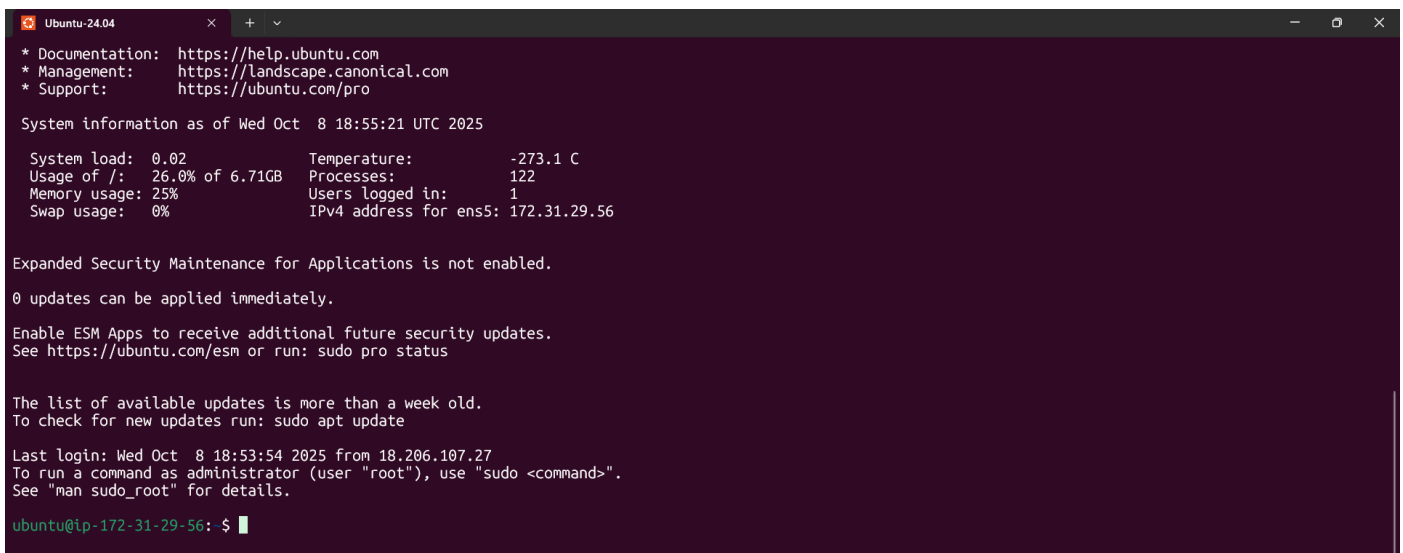
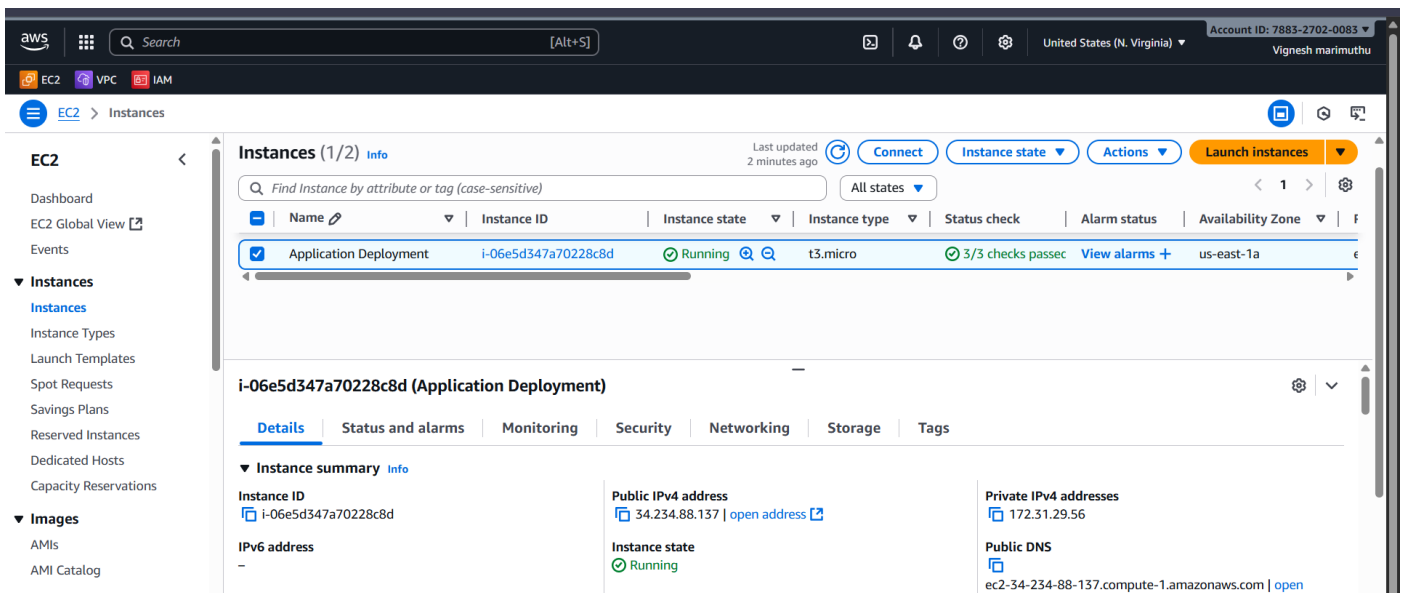
- Source: GitHub
- Build: AWS CodeBuild project
- Deploy: AWS CodeDeploy or deploy to EKS via Lambda or custom script.

Monitoring:

- Use CloudWatch Logs to track build, deploy, and application logs.

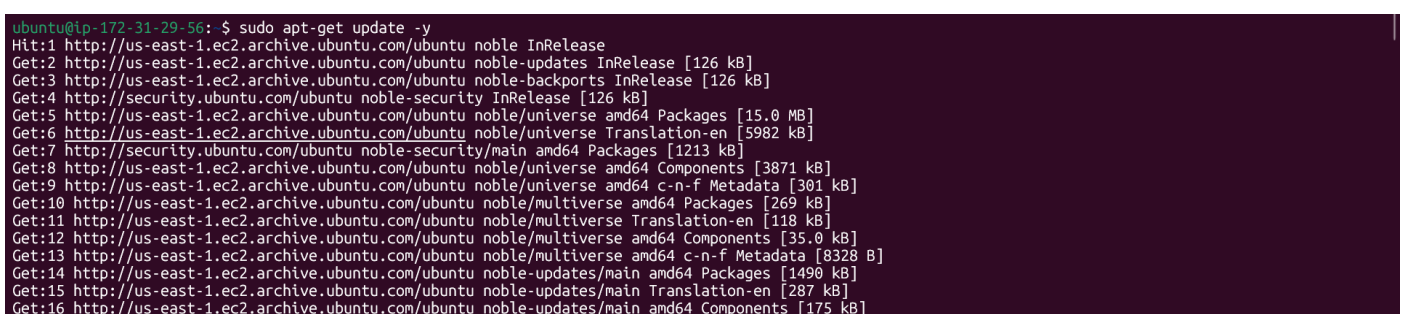
Solution:

✓ 1. Launched an EC2 instance and connected to Ubuntu via SSH



- This EC2 acts as **DevOps control machine** to manage AWS resources, build images, and run **kubectl** commands.

✓ 2. Updated and Installed Required Tools



```
ubuntu@ip-172-31-29-56:~$ sudo apt-get install -y docker.io
sudo usermod -aG docker $USER
newgrp docker
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc rinse zfs-fuse | zfsutils
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan
0 upgraded, 8 newly installed, 0 to remove and 41 not upgraded.
Need to get 75.9 MB of archives.
After this operation, 288 MB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 bridge-utils amd64 1.7.1-1ubuntu2 [33.9 kB]
```

```
You can now run: /usr/local/bin/aws --version
aws-cli/2.31.10 Python/3.13.7 Linux/6.14.0-1011-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-29-56:~$ curl -LO "https://dl.k8s.io/release/stable.txt)(curl -Ls https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
chmod +x kubectl && sudo mv kubectl /usr/local/bin/
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 138 100 138 0 0 2082 0 --:--:-- --:--:-- --:--:-- 2090
100 57.7M 100 57.7M 0 0 233M 0 --:--:-- --:--:-- --:--:-- 456M
ubuntu@ip-172-31-29-56:~$ curl -sSL "https://github.com/eksctl-io/eksctl/releases/latest/download/eksctl_$(uname -s)_amd64.tar.gz" | sudo tar xz -C /usr/local/bin
ubuntu@ip-172-31-29-56:~$
```

```
ubuntu@ip-172-31-29-56:~$ aws --version
kubectl version --client=true --output=yaml
eksctl version
aws-cli/2.31.10 Python/3.13.7 Linux/6.14.0-1011-aws exe/x86_64.ubuntu.24
clientVersion:
  buildDate: "2025-09-09T19:44:50Z"
  compiler: gc
  gitCommit: 93248f9ae092f571eb870b7664c534bfc7d00f03
  gitTreeState: clean
  gitVersion: v1.34.1
  goVersion: go1.24.6
  major: "1"
  minor: "34"
  platform: linux/amd64
kustomizeVersion: v5.7.1
0.215.0
ubuntu@ip-172-31-29-56:~$
```

- Installed all tools required to build, push, and deploy the app — like Docker, AWS CLI, kubectl, eksctl.

✓ 3. Configure AWS CLI

```
ubuntu@ip-172-31-29-56:~$ aws configure
AWS Access Key ID [None]: AKIA3PC7TWYZVJBMRPE7
AWS Secret Access Key [None]: l3pKlNg1CvpsSVMH1hBTGyLbfJisaz0ZC5JVgHEi
Default region name [None]: us-east-1
Default output format [None]: json
ubuntu@ip-172-31-29-56:~$
```

- To allow your EC2 to access AWS services like ECR, EKS, and CodeBuild.

✓ 4. Created an ECR Repository

```
ubuntu@ip-172-31-29-56:~$ aws ecr create-repository --repository-name brain-tasks-app --region us-east-1
{
  "repository": {
    "repositoryArn": "arn:aws:ecr:us-east-1:788327020083:repository/brain-tasks-app",
    "registryId": "788327020083",
    "repositoryName": "brain-tasks-app",
    "repositoryUri": "788327020083.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app",
    "createdAt": "2025-10-08T19:00:45.314000+00:00",
    "imageTagMutability": "MUTABLE",
    "imageScanningConfiguration": {
      "scanOnPush": false
    },
    "encryptionConfiguration": {
      "encryptionType": "AES256"
    }
  }
}
ubuntu@ip-172-31-29-56:~$
```

- Created an AWS ECR repo to store the Docker image.
- ECR (Elastic Container Registry) acts as a **private** container image storage.

✓ 5. Cloned the React App Repository & Push it to My Repository

```
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ git push origin main
Username for 'https://github.com': vigneshmarimuthu
Password for 'https://vigneshmarimuthu@github.com':
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 2 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (8/8), 100.04 KiB | 100.04 MiB/s, done.
Total 8 (delta 0), reused 8 (delta 0), pack-reused 0
To https://github.com/vigneshmarimuthu/brain-tasks-app-v2.git
 * [new branch]    main -> main
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$
```

- To get the application code for Dockerizing and deploying.
- My Repository Link - <https://github.com/vigneshmarimuthu/brain-tasks-app-v2>

✓ 6. Created Dockerfile

```
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ nano Dockerfile
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ cat Dockerfile
# Use NGINX image
FROM public.ecr.aws/nginx/nginx:alpine

# Copy built static files
COPY dist/ /usr/share/nginx/html

# Expose port 80
EXPOSE 80

# Start nginx
CMD ["nginx", "-g", "daemon off;"]
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$
```

- Used NGINX to serve the React build output (dist folder) as static files.

✓ 7. Build and Push Image to ECR

```
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ ACCOUNT_ID=$(aws sts get-caller-identity --query Account --output text)
REGION=us-east-1
REPO=brain-tasks-app
ECR_URI=$ACCOUNT_ID.dkr.ecr.$REGION.amazonaws.com/$REPO
IMAGE_TAG=v1
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ aws ecr get-login-password --region $REGION | docker login --username AWS --password-stdin $ACCOUNT_ID.dkr.ecr.$REGION.amazonaws.com

WARNING! Your credentials are stored unencrypted in '/home/ubuntu/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ docker build -t $REPO:$IMAGE_TAG .
docker tag $REPO:$IMAGE_TAG $ECR_URI:$IMAGE_TAG
docker push $ECR_URI:$IMAGE_TAG
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 487.4kB
Step 1/4 : FROM public.ecr.aws/nginx/nginx:alpine
alpine: Pulling from nginx/nginx
9824c27679d3: Pulling fs layer
6bc572a340ec: Pulling fs layer
403e3f251637: Pulling fs layer
```

```
Successfully built d98302afe19d
Successfully tagged brain-tasks-app:v1
The push refers to repository [788327020083.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app]
4416987ac656: Pushed
f9985d3fc94d: Pushed
d208138be39d: Pushed
a2b76470e8f1: Pushed
917b2c97271e: Pushed
16ca725632e5: Pushed
7978a9c91f72: Pushed
b6ff0212304e: Pushed
418dccb7d85a: Pushed
v1: digest: sha256:d59dfa657f9e6de290dfa933b8a21a7281a0e43fa1e404159226bdf979763cc size: 2199
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$
```

- To make the image accessible by EKS cluster.

✓ 8. Created EKS Cluster

```
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ eksctl create cluster --name brain-cluster --region us-east-1 --nodes 2
aws eks update-kubeconfig --name brain-cluster --region us-east-1
kubectl get nodes
2025-10-08 19:08:43 [i] eksctl version 0.215.0
2025-10-08 19:08:43 [i] using region us-east-1
2025-10-08 19:08:43 [i] setting availability zones to [us-east-1c us-east-1b]
2025-10-08 19:08:43 [i] subnets for us-east-1c - public:192.168.0.0/19 private:192.168.64.0/19
2025-10-08 19:08:43 [i] subnets for us-east-1b - public:192.168.32.0/19 private:192.168.96.0/19
2025-10-08 19:08:43 [i] nodegroup "ng-92c7f4ca" will use "" [AmazonLinux2023/1.32]
2025-10-08 19:08:43 [!] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-on
s will no longer be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn mo
re: https://eksctl.io/usage/auto-mode/
2025-10-08 19:08:43 [i] using Kubernetes version 1.32
2025-10-08 19:08:43 [i] creating EKS cluster "brain-cluster" in "us-east-1" region with managed nodes
2025-10-08 19:08:43 [i] will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2025-10-08 19:08:43 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=brain-c
luster'
```

```
2025-10-08 19:22:57 [i] created 1 managed nodegroup(s) in cluster "brain-cluster"
2025-10-08 19:22:58 [i] kubectl command should work with "/home/ubuntu/.kube/config", try 'kubectl get nodes'
2025-10-08 19:22:58 [i] EKS cluster "brain-cluster" in "us-east-1" region is ready
Added new context arn:aws:eks:us-east-1:788327020083:cluster/brain-cluster to /home/ubuntu/.kube/config
NAME STATUS ROLES AGE VERSION
ip-192-168-10-197.ec2.internal Ready <none> 103s v1.32.9-eks-113cf36
ip-192-168-46-233.ec2.internal Ready <none> 102s v1.32.9-eks-113cf36
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$
```

- EKS is **Kubernetes cluster** where the container will run.

✓ 9. Created Kubernetes Deployment and Service

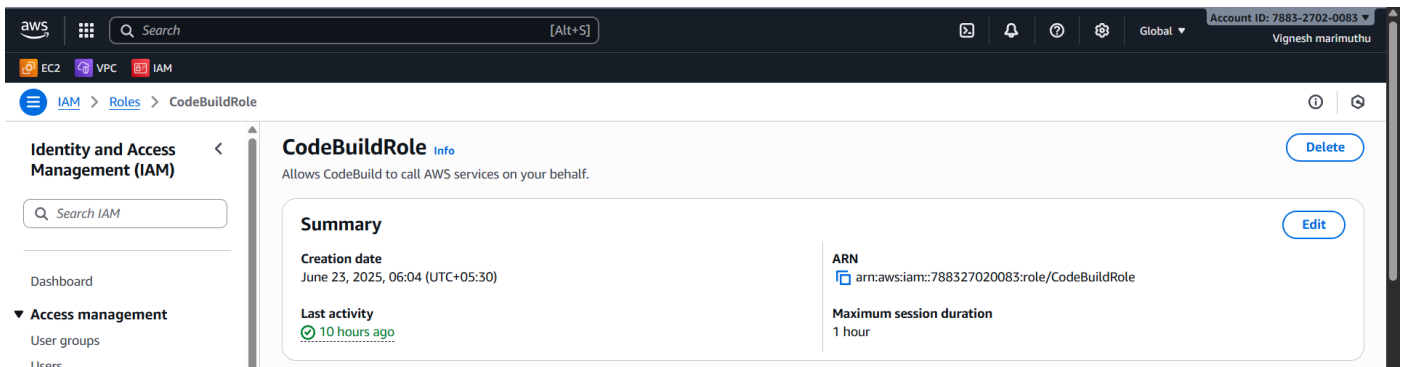
```
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ nano deployment.yaml
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-deploy
  labels:
    app: brain
spec:
  replicas: 2
  selector:
    matchLabels:
      app: brain
  template:
    metadata:
      labels:
        app: brain
    spec:
      containers:
        - name: brain-container
          image: <ACCOUNT_ID>.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:v1
          ports:
            - containerPort: 80
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$
```

```
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ nano service.yaml
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: brain-service
spec:
  type: LoadBalancer
  selector:
    app: brain
  ports:
    - port: 80
      targetPort: 80
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$
```

```
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
kubectl get svc brain-service -o wide
deployment.apps/brain-deploy created
service/brain-service created
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE SELECTOR
brain-service LoadBalancer 10.100.205.12 <pending> 80:30449/TCP 1s app=brain
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$
```

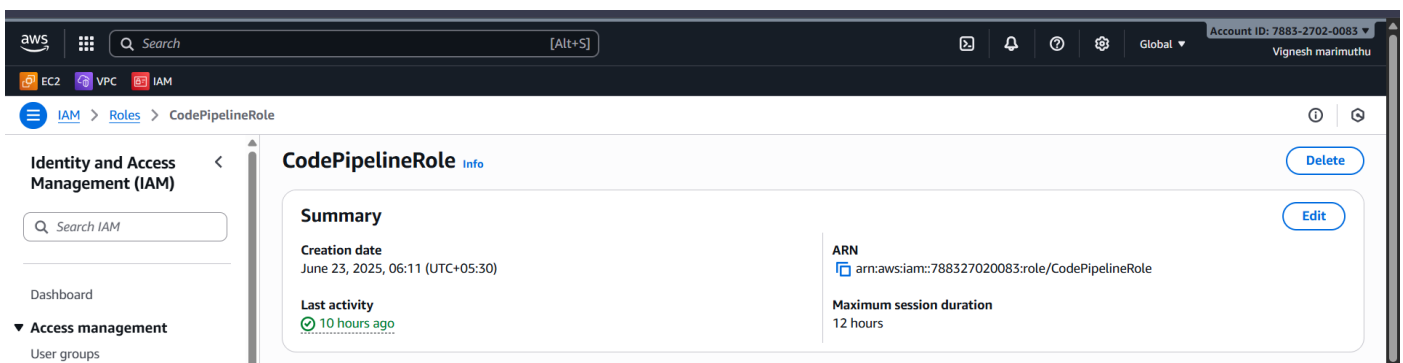
- To deploy the app on the EKS cluster using Deployment and Service.

✓ 9. Created IAM Roles for CI/CD



✓ Policies:

- AmazonEC2ContainerRegistryPowerUser
- CloudWatchLogsFullAccess
- AmazonS3FullAccess



✓ Policies:

- AWSCodePipelineFullAccess
 - CloudWatchLogsFullAccess
 - AmazonS3FullAccess
 - AWSCodeBuildDeveloperAccess
- To give CodeBuild and CodePipeline permission to build and deploy automatically.

✓ 10. Created buildspec.yml

```
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ nano buildspec.yml
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ cat buildspec.yml
version: 0.2

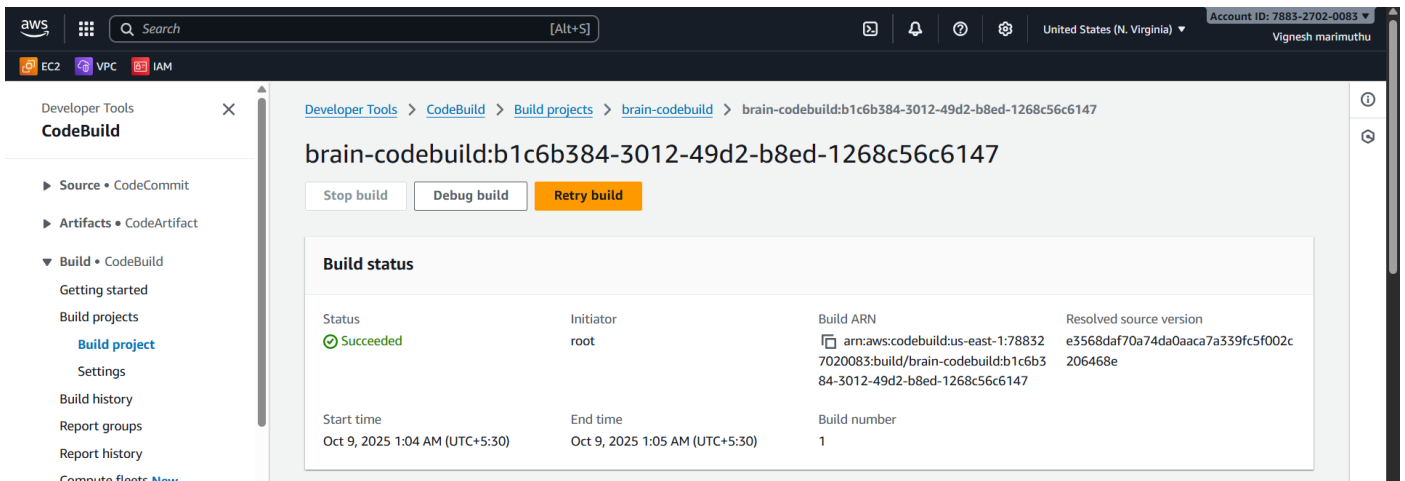
phases:
  install:
    commands:
      - echo "Skipping npm install because only static files exist."
  build:
    commands:
      - echo "No build step required. Using prebuilt dist/ files."
artifacts:
  files:
    - '**/*'
  base-directory: dist
  discard-paths: no
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$
```

- Define how CodeBuild builds and deploys the image.

✓ 11. Created CodeBuild Project

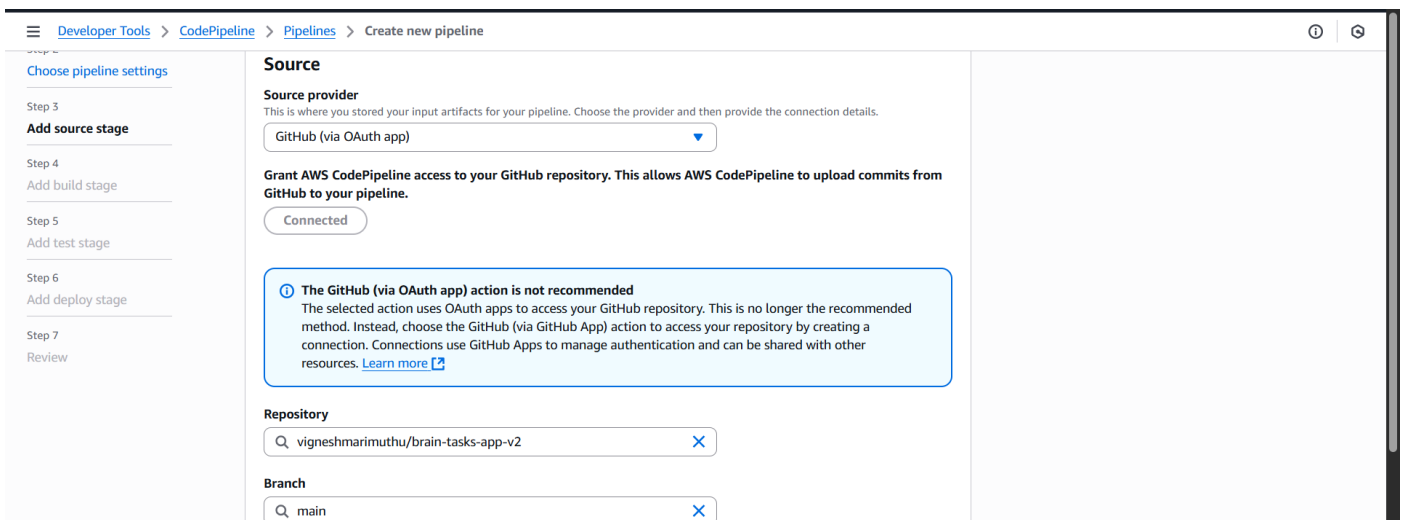
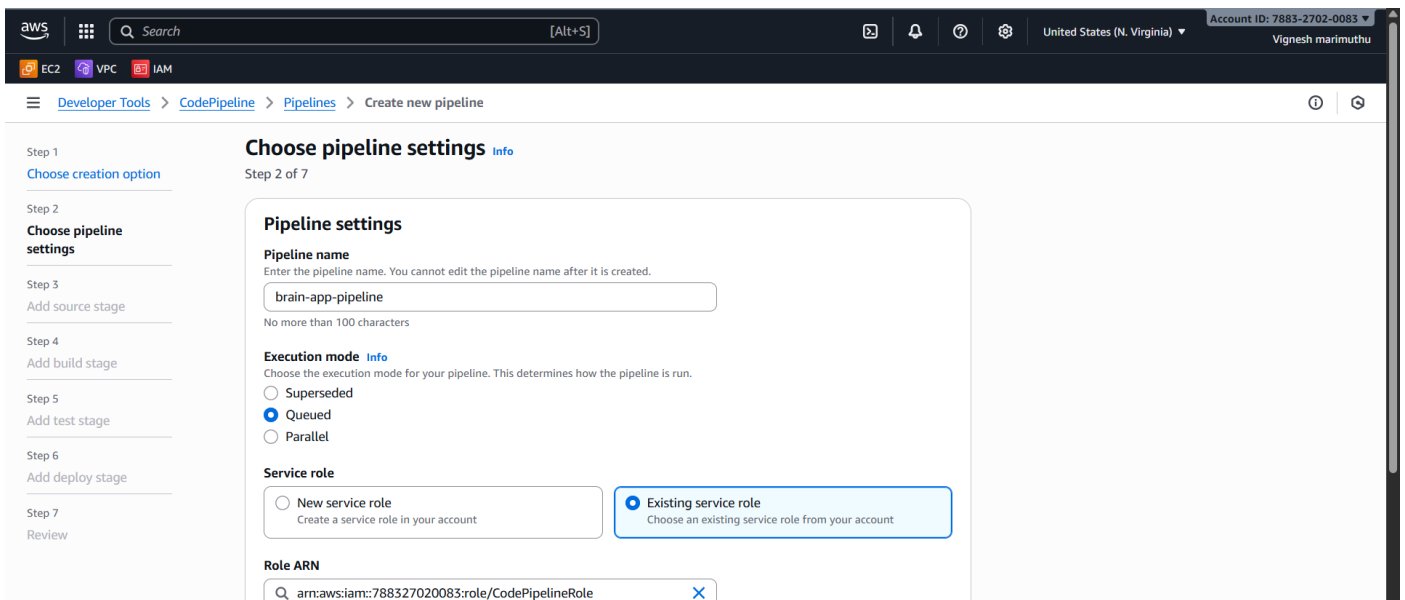
The screenshot shows the AWS CodeBuild 'Create build project' console page. The breadcrumb navigation is 'Developer Tools > CodeBuild > Build projects > Create build project'. The page title is 'Create build project'. Under 'Project configuration', the 'Project name' field contains 'brain-codebuild'. Below it, a note states: 'A project name must be 2 to 255 characters. It can include the letters A-Z and a-z, the numbers 0-9, and the special characters - and _.' The 'Project type' section has two options: 'Default project' (selected with a radio button) and 'Runner project'. The 'Default project' description is 'Create a custom CodeBuild project.' The 'Runner project' description is 'Create a CodeBuild managed runner for workflows in GitHub Actions, GitHub Enterprise Actions, GitLab, or Buildkite.' At the bottom, there is an 'Additional configuration' section with a description: 'Description, public build access, build badge, concurrent build limit, tags'.

The screenshot shows the 'Source' configuration section of the AWS CodeBuild console. It features an 'Add source' button. Under 'Source 1 - Primary', the 'Source provider' is set to 'GitHub'. A green checkmark indicates 'Your account is successfully connected by using an AWS managed GitHub App. Manage account credentials.' There is an unchecked checkbox for 'Use override credentials for this project only'. The 'Repository' section has three options: 'Repository in my GitHub account' (unchecked), 'Public repository' (selected with a radio button), and 'GitHub scoped webhook' (unchecked). The 'Repository URL' field contains 'https://github.com/vigneshmarimuthu/brain-tasks-app-v2', with a placeholder text 'https://github.com/<user-name>/<repository-name>' below it.



- To automate build & push image to ECR.
- Build succeeds and image pushed to ECR.

✓ 12. Created CodePipeline



Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1
Choose creation option

Step 2
Choose pipeline settings

Step 3
Add source stage

Step 4
Add build stage

Step 5
Add test stage

Step 6
Add deploy stage

Step 7
Review

Add build stage Info

Step 4 of 7

Build - *optional*

Build provider
Choose the tool you want to use to run build commands and specify artifacts for your build action.

☐ Commands
 ☒ Other build providers

AWS CodeBuild

Project name
Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.

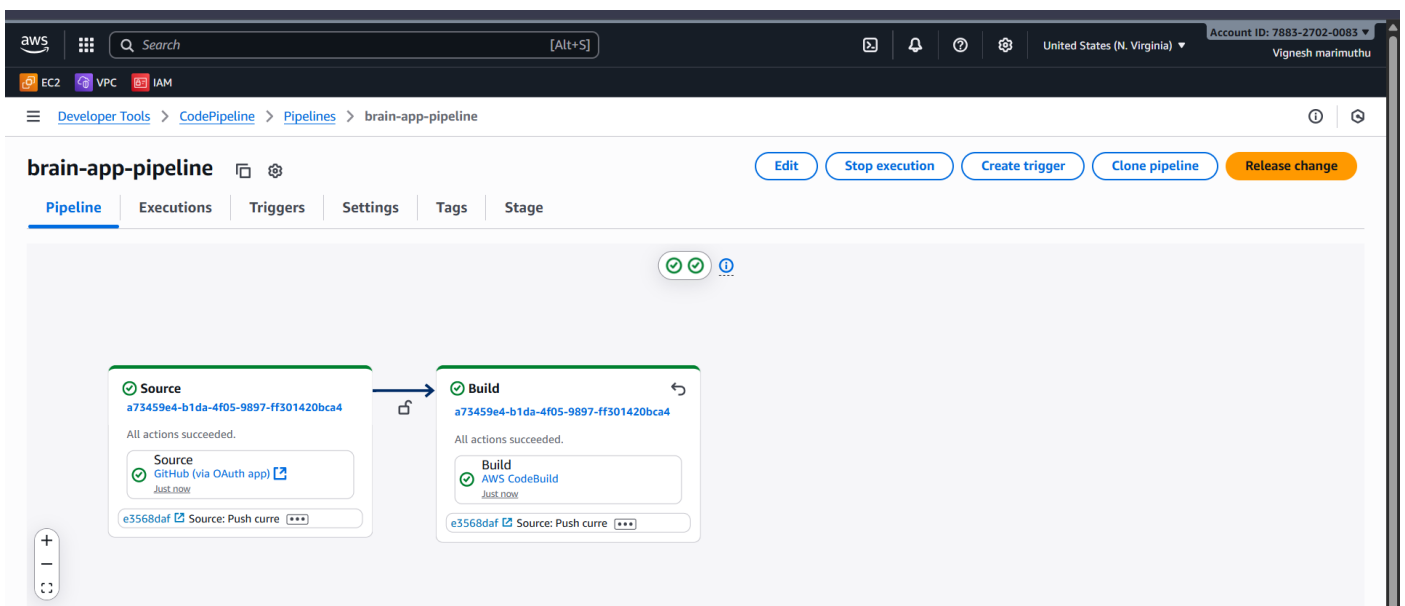
or [Create project](#)

☐ Define buildspec override - *optional*
Buildspec file or definition that overrides the latest one defined in the build project, for this build only.

```

4 files changed, 57 insertions(+)
create mode 100644 Dockerfile
create mode 100644 buildspec.yaml
create mode 100644 deployment.yaml
create mode 100644 service.yaml
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ git push origin main
Username for 'https://github.com': vigneshmarimuthu
Password for 'https://vigneshmarimuthu@github.com':
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.08 KiB | 1.08 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/vigneshmarimuthu/brain-tasks-app-v2.git
 b7d121f..e3568da  main -> main

```

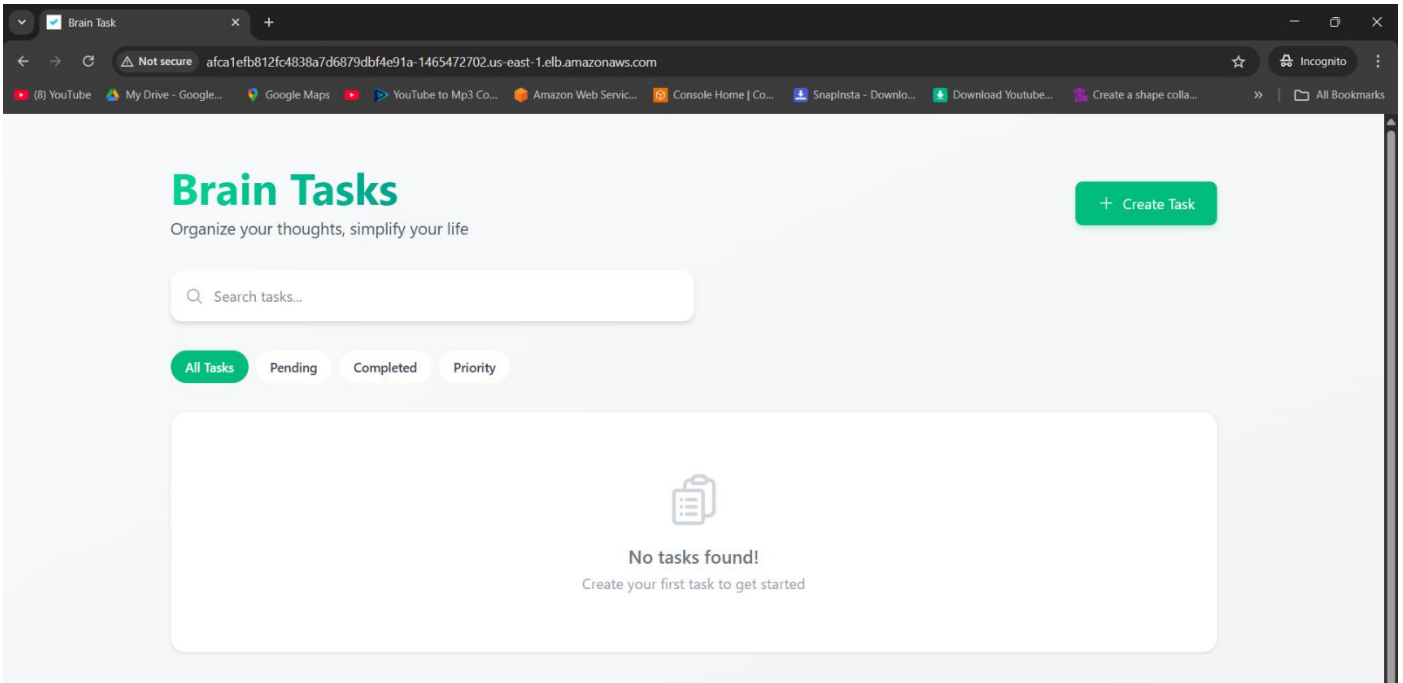


- To automate deployment from GitHub to EKS.
- Pipeline runs automatically when i push to GitHub.

✓ 9. Get LoadBalancer DNS

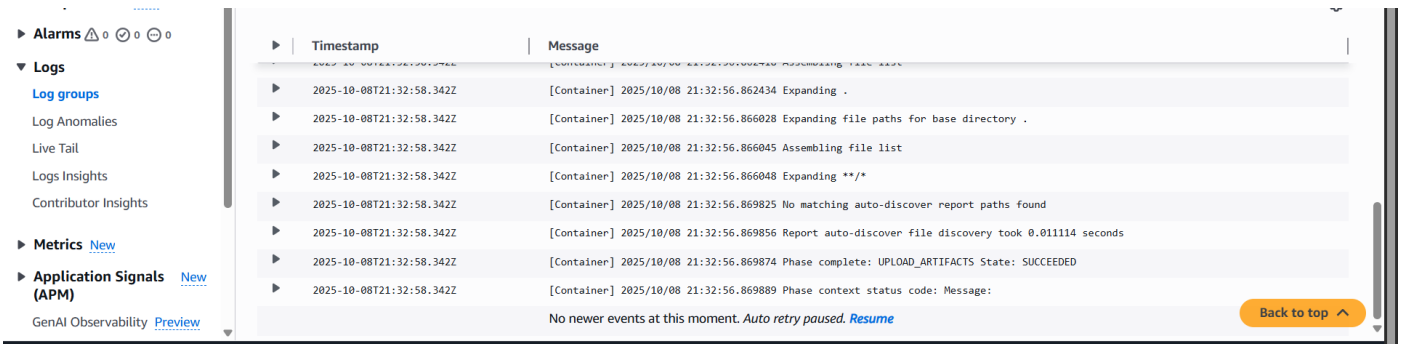
```
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
brain-deploy-649c94bd4c-f8hks       1/1     Running   0           16s
brain-deploy-649c94bd4c-mccgq       1/1     Running   0           13s
```

```
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$ kubectl get svc brain-service -o jsonpath='{.status.loadBalancer.ingress[0].hostname}'; echo
afca1efb812fc4838a7d6879dbf4e91a-1465472702.us-east-1.elb.amazonaws.com
ubuntu@ip-172-31-29-56:~/brain-tasks-app-v2$
```



- The DNS has been tested and confirmed to be working.

✓ 9. Monitoring with Amazon CloudWatch



- To track build, deploy, and app logs.